

University of Arkansas STAT6393V - Week 5

Week 5 - Final Project: Estimation and Uncertainty Quantification of rainfall data.

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Introduction

All models either work for discrete or continuous data, but not both. The challenge in modeling discrete and continuous data is to find a model that can capture the underlying structure of the data while also being able to quantify the uncertainty in the estimates. In this project, we will explore different models for estimating rainfall data and quantifying the uncertainty in the estimates.

The challenge in modeling discrete and continuous data is to find a model that can capture the underlying structure of the data while also being able to quantify the uncertainty in the estimates. In this project, we will explore different models for estimating rainfall data and quantifying the uncertainty in the estimates.

Part 1: Setup

1. Data Exploration and Preparation - Load the dataset:

```
weather <- readRDS("weather.RDS")  
dim(weather)
```

```
[1] 584000      7
```

```
str(weather)
```

```
'data.frame':  584000 obs. of  7 variables:  
 $ time : chr  "1951-01-01 00:00:00" "1951-01-01 00:00:00" "1951-01-01 00:00:00" "1951-01-01  
 $ lat  : num  34.4 34.4 34.4 34.4 34.4 ...  
 $ lon  : num  249 250 251 252 254 ...  
 $ tmax : num  277 277 277 280 281 ...  
 $ pr   : num  7.816 9.494 5.287 3.324 0.139 ...  
 $ month: num   1 1 1 1 1 1 1 1 1 ...  
 $ loc  : Factor w/ 25 levels "1","2","3","4",...: 1 2 3 4 5 6 7 8 9 10 ...
```

```
max <- max(weather$time)
min <- min(weather$time)
```

```
max
```

```
[1] "2014-12-31 00:00:00"
```

```
min
```

```
[1] "1951-01-01 00:00:00"
```

The weather dataset contains daily weather data from january, 1, 1951 to december 31, 2014, comprising 7 variables and 584000 observations. The variables are: time, latitude, longitude, maximum temperature, precipitation, month, and location.. The dataset is in long format, with each row representing a single observation for a specific location and time.

- Select a subset of the dataset:

```
data <- weather[weather$month == 6, ]
dim(data)
```

```
[1] 48000      7
```

```
head(data)
```

		time	lat	lon	tmax	pr	month	loc
3776	1951-06-01	00:00:00	34.39791	248.75	296.5269	0	6	1
3777	1951-06-01	00:00:00	34.39791	250.00	295.7346	0	6	2
3778	1951-06-01	00:00:00	34.39791	251.25	296.4243	0	6	3
3779	1951-06-01	00:00:00	34.39791	252.50	299.7874	0	6	4
3780	1951-06-01	00:00:00	34.39791	253.75	301.7009	0	6	5
3781	1951-06-01	00:00:00	35.34031	248.75	296.5229	0	6	6

- Set all precipitation values less than 0.01 to 0:

```
data$pr[data$pr < 0.01] <- 0
str(data)
```

```
'data.frame':  48000 obs. of  7 variables:
 $ time : chr  "1951-06-01 00:00:00" "1951-06-01 00:00:00" "1951-06-01 00:00:00" "1951-06-01
 $ lat  : num  34.4 34.4 34.4 34.4 34.4 ...
 $ lon  : num  249 250 251 252 254 ...
 $ tmax : num  297 296 296 300 302 ...
 $ pr   : num  0 0 0 0 0 0 0 0 0 0 ...
 $ month: num  6 6 6 6 6 6 6 6 6 6 ...
 $ loc  : Factor w/ 25 levels "1","2","3","4",...: 1 2 3 4 5 6 7 8 9 10 ...
```

In this project, we will only focus on the month of June. The dataset contains 48000 observations for the month of June, comprising 7 variables.

- Split the data into training and testing sets:

```
set.seed(123)
train_rows <- which(year(data$time) > 1960 & year(data$time) < 2001)
train_set <- data[train_rows, ]
test_set <- data[-train_rows, ]

dim(train_set)
```

```
[1] 30000      7
```

```
head(train_set)
```

		time	lat	lon	tmax	pr	month	loc
95026	1961-06-01	00:00:00	34.39791	248.75	297.8128	0	6	1
95027	1961-06-01	00:00:00	34.39791	250.00	298.3752	0	6	2
95028	1961-06-01	00:00:00	34.39791	251.25	298.8843	0	6	3
95029	1961-06-01	00:00:00	34.39791	252.50	300.9040	0	6	4
95030	1961-06-01	00:00:00	34.39791	253.75	302.8736	0	6	5
95031	1961-06-01	00:00:00	35.34031	248.75	298.5822	0	6	6

```
dim(test_set)
```

```
[1] 18000      7
```

```
head(test_set)
```

		time	lat	lon	tmax	pr	month	loc
3776	1951-06-01	00:00:00	34.39791	248.75	296.5269	0	6	1
3777	1951-06-01	00:00:00	34.39791	250.00	295.7346	0	6	2
3778	1951-06-01	00:00:00	34.39791	251.25	296.4243	0	6	3
3779	1951-06-01	00:00:00	34.39791	252.50	299.7874	0	6	4
3780	1951-06-01	00:00:00	34.39791	253.75	301.7009	0	6	5
3781	1951-06-01	00:00:00	35.34031	248.75	296.5229	0	6	6

If we split the data into training and testing sets, we will use the data from 1961 to 2000 as the training set and the data from 1951 to 1960 and 2001 to 2014 as the testing set. The training set contains 30000 observations, while the testing set contains 18000 observations.

2. Custom Loss Function for semi-continuous gamma mixture:

```
# Custom loss function for semi-continuous gamma mixture
GSCM_loss <- function(y_true, y_pred) {
  # Extract the parameters from the predicted values
  pi <- y_pred[, 1]
  alpha <- y_pred[, 2]
  beta <- y_pred[, 3]
}
```

```
GSCMM_nll <- function(pred,true,...){
  yobs <- true[,1, drop = FALSE] # Extract Observed precipitation values
  rainy <- ifelse(yobs > 0, 1, 0) # Create a binary variable indicating
  ↪ whether it rained or not

  pi0 <- torch_sigmoid(pred[, 1, drop = FALSE]) # Extract the predicted
  ↪ probability of no rain
  alpha <- torch_exp(pred[, 2, drop = FALSE]) + 1e-6 # Extract the predicted
  ↪ shape parameter for the gamma distribution
  beta <- torch_exp(pred[, 3, drop = FALSE]) + 1e-6 # Extract the predicted
  ↪ scale parameter for the gamma distribution

  alpha <- alpha[rainy == 1] # Keep only the shape parameters for the rainy
  ↪ days
  beta <- beta[rainy == 1] # Keep only the scale parameters for the rainy
  ↪ days

  #Yoinking from slides:
  ll <- sum(torch_log(pi0[rainy == 0])) +
    sum(torch_log(1 - pi0[rainy == 1]) +
      ↪ torch::distr_gamma(alpha,beta)$log_prob(yobs[rainy == 1])) #
      ↪ Calculate the log-likelihood for the semi-continuous gamma
      ↪ mixture model

  -ll
}
```

```
GSCMM_nll <- function(pred,true,...){
  yobs <- true[,1, drop = FALSE]
  rainy <- ifelse(yobs > 0, 1, 0)

  pi0 <- torch_sigmoid(pred[, 1, drop = FALSE])
  alpha <- torch_exp(pred[, 2, drop = FALSE]) + 1e-6
  beta <- torch_exp(pred[, 3, drop = FALSE]) + 1e-6

  alpha <- alpha[rainy == 1]
  beta <- beta[rainy == 1]

  #Yoinking from slides:
  ll <- sum(torch_log(pi0[rainy == 0])) +
    sum(torch_log(1 - pi0[rainy == 1]) +
      ↪ torch::distr_gamma(alpha,beta)$log_prob(yobs[rainy == 1]))

  -ll
}
```

i Part 2: Model Fitting

1. Fitting a model for pr | tmax:

- Pick a location and subset the data for that location. In this case, we will pick location 1.

```
train_loc1 <- train_set[train_set$loc == 1, ]
test_loc1 <- test_set[test_set$loc == 1, ]
dim(train_loc1)
```

```
[1] 1200    7
```

```
dim(test_loc1)
```

```
[1] 720    7
```

```
train_df <- data.frame(x = train_loc1$tmax, y = train_loc1$pr)
test_df <- data.frame(x = test_loc1$tmax, y = test_loc1$pr)
```

```
#I just picked location 1, y'all can change it up if you want!
loc1 <- which(train_set$loc == 1)

loc1train <- train_set[loc1,]
ytrain <- loc1train$pr
xtrain <- loc1train$tmax

loc1test <- test_set[loc1,]

ytest <- loc1train$pr
xtest <- loc1train$tmax

train_df <- data.frame(x = xtrain, y = ytrain)
test_df <- data.frame(x = xtest, y = ytest)
```

```
# | fig-high: 8
# | fig-width: 6
par(mfrow = c(1, 2))
model1 <- cito::dnn(
  cbind(y,y,y) ~ x, data = train_df, loss = GSCMM_nll,
  hidden = c(50, 50), activation = "relu",
  epochs = 50, lr = 0.001, batchsize = 100,
  optimizer = "adam", validation = 0.2,
  verbose = F, plot = T
)
```

Warning: the 'nobars' function has moved to the reformulas package. Please update your imports.
This warning is displayed once per session.

```
model2 <- cito::dnn(
  cbind(y,y,y) ~ x, data = train_df, loss = GSCMM_nll,
  hidden = c(100, 100), activation = "relu",
  epochs = 50, lr = 0.001, batchsize = 100,
  optimizer = "adam", validation = 0.2,
  verbose = F, plot = T
)
```

Part 3: Model Validation

Coming Soon!

Part 4: Scientific Questions

Coming Soon!

i Appendix

Coming Soon!